Name: Kanika Saini

SAP ID: 590034098

Program: B.Tech CSE (B16)

Question 1 :

Problem Statement-

Write a C program to find the hypotenuse, area, and perimeter of a right-angled triangle when its base and height are given.

Step 1: Input, Output and Formula

Input : Base (b) , Height (h)

Output : Hypotenuse , Area , Perimeter

Formula: Hypotenuse = $\sqrt{(b^2 + h^2)}$

Area = $(1/2) \times b \times h$

Perimeter = b + h + Hypotenuse

Step 2: Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
1.	Base = 3 Height = 4	Hypotenuse = 5 , Area = 6, Perimeter=12	Same	Pass
2.	Base = 5 Height = 12	Hypotenuse = 13, Area = 30, Perimeter = 30	Same	Pass
3.	Base = 8 Height = 15	Hypotenuse = 17, Area = 60, Perimeter = 40	Same	Pass

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Step 3: Algorithm-

1. Start.
2. Read the base and height.
3. Calculate the hypotenuse using the Pythagoras theorem.
4. Calculate the area.
5. Calculate the perimeter.
6. Display the hypotenuse, area, and perimeter.
7. Stop.

Algorithm Evaluation-

The algorithm accepts the base and height as input. It applies the required mathematical formulas to calculate the hypotenuse, area, and perimeter. The algorithm is simple, accurate, and produces the correct output.

Step 4: C Program

Make ‘triangle.c’ file first then write this code-

```
#include<stdio.h>
#include<math.h>

int main()
{
	float base, height, hypotenuse, area, perimeter;
	printf("enter base: ");
	scanf("%f",&base);
	printf("enter height: ");
	scanf("%f",&height);

	hypotenuse =sqrt(base*base+height*height);
	area= 0.5*base*height;
	perimeter= base+height+hypotenuse;

	printf("Hypotenuse = %.2f\n",hypotenuse);
	printf("Area = %.2f\n",area);
```

```
    printf("Perimeter = %.2f\n",perimeter);

    return 0;
}
```

Step 5: Compilation and Execution

Compile the program using: “ cc triangle.c -lm ”

Execute using: “ ./a.out ”

```
┌──(kali㉿kali)-[~/Desktop]
└─$ cc triangle.c -lm

┌──(kali㉿kali)-[~/Desktop]
└─$ ./a.out
enter base: 3
enter height: 4
Hypotenuse = 5.00
Area = 6.00
Perimeter = 12.00
```

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./a.out
enter base: 5
enter height: 12
Hypotenuse = 13.00
Area = 30.00
Perimeter = 30.00
```

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./a.out
enter base: 8
enter height: 15
Hypotenuse = 17.00
Area = 60.00
Perimeter = 40.00
```

Code Evaluation-

The program compiled successfully without any errors. It correctly calculated the hypotenuse, area, and perimeter for all test cases. The actual output matched the expected output.

Conclusion-

The program was successfully compiled and executed. All the test cases passed successfully. Hence, the program correctly calculates the hypotenuse, area, and perimeter of a right-angled triangle.

Question 2:

Problem Statement-

Write a C program to find the apothem, perimeter, and each interior angle of a regular polygon when its number of sides, side length, and area are given.

Step 1: Input, Output and Formula

<u>Input</u>: Number of sides (n) , Side length (s), Area (A)

<u>Output</u>: Apothem, Perimeter, Interior angle

<u>Formula</u> : Perimeter = $n \times s$, Apothem = $(2 \times \text{Area}) / \text{Perimeter}$, Interior Angle = $((n - 2) \times 180) / n$

Step 2: Test Case Table

Test Case	**Input**	**Expected Output**	**Actual Output**	**Result**
1.	n=6, s=5, Area=64.95	Perimeter=30, Apothem=4.33, Interior Angle=120(degree)	Same	Pass
2.	n=4, s=8, Area=64	Perimeter=32, Apothem=4, Interior Angle=90(degree)	Same	Pass
3.	n=5, s=6, Area=61.94	Perimeter=30, Apothem=4.13, Interior Angle=108(degree)	Same	Pass

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Step 3: Algorithm

1. Start.
2. Read the number of sides, side length, and area.
3. Calculate the perimeter.
4. Calculate the apothem.
5. Calculate the interior angle.
6. Display the perimeter, apothem, and interior angle.
7. Stop.

Algorithm Evaluation-

The algorithm accepts the required inputs and calculates the perimeter, apothem, and interior angle using the given formulas. It is easy to understand and provides accurate results.

Step 4: C Program

Make ‘polygon.c’ file first then write this code-

```c
#include<stdio.h>
int main()
{
    int n;
    float side, area, apothem, perimeter, angle;
    printf("Enter number of sides: ");
    scanf("%d", &n);
    printf("Enter side length: ");
    scanf("%f",&side);
    printf("Enter area: ");
    scanf("%f", &area);

    perimeter = n*side;
    apothem = (2* area)/perimeter;
    angle = ((n-2)*180.0)/n;

    printf("Perimeter = %.2f\n", perimeter);
    printf("Apothem = %.2f\n", apothem);
    printf("Interior Angle = %.2f degrees\n", angle);
```

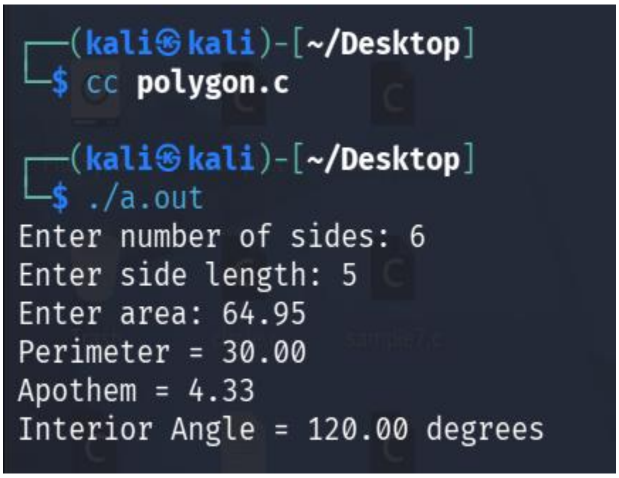
return 0;

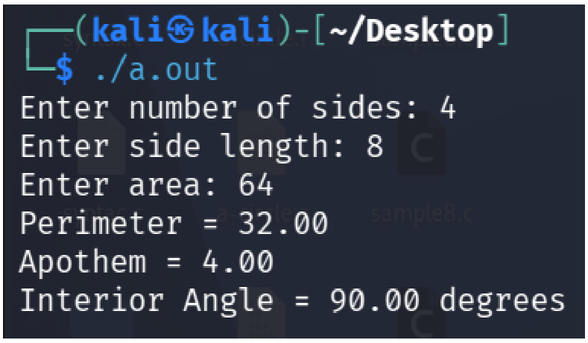
}

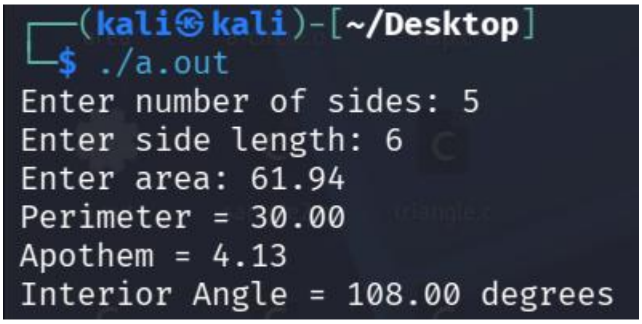
Step 5: Compilation and Execution

Compile the program using: “cc polygon.c”

Execute using: “./a.out”







Code Evaluation-

The program compiled and executed successfully. It correctly calculated the perimeter, apothem, and interior angle for all test cases. The expected and actual outputs matched.

Conclusion-

The program was compiled and executed successfully. The results obtained from all test cases were correct. Therefore, the program satisfies the given problem statement and performs the required calculations accurately.